



Oral Surgery

Third stage

Dr. Mohammed Hassan
B.D.S., M.Sc., Ph.D. (Oral Medicine).

Complications of Exodontia (part 2)

7- Excessive bleeding after extraction

☐ At the beginning one must understand that some **slight oozing of blood** for several hours following tooth extraction is considered normal, but sometime **excessive or abnormal bleeding** may occur following tooth extraction.

☐ The causes of excessive bleeding may be:

A. Local factors:

The local causes which are the commonest causes for prolonged bleeding, due to:

1. Gross tissue damage, when there is severe bone injury and tearing of the periosteum many vessels are opened, also severe gingival lacerations.
2. Damage to **large arteries** like inferior dental vessel or greater palatine vessels may lead to profuse bleeding.
3. **The presence of tumors especially hemangioma** and other vascular abnormalities may lead to such complication.
4. **Post-operative infection** of the extraction wound causing erosion of the blood vessel leading to secondary haemorrhage.

B systemic factors:

Like thrombocytopenia, reduction in the clotting factors, anticoagulant medications, liver disease and hypertension all these factors may lead to severe bleeding; so good history and clinical examination and blood investigation is very important and essential before any extraction especially if the patient had a history of bleeding on previous extractions or trauma.

o-Damage to the surrounding soft tissues:

- a. Damage to the lower lip.
- The lip may be pressed or crushed between the handles of the forceps on extraction of upper teeth if insufficient care is not taken.
- b Damage to the tongue and floor of the mouth
- Slipping of elevator during extraction may lead to damage or wounds in the floor of the oral cavity, there are many vital structures in the floor of the oral cavity which might be damage like [sublingual gland submandibular duct, lingual nerve & tongue.
- So the operator should always keep in his mind that supporting of elevator during extraction is very important.

■ LIP INJURY

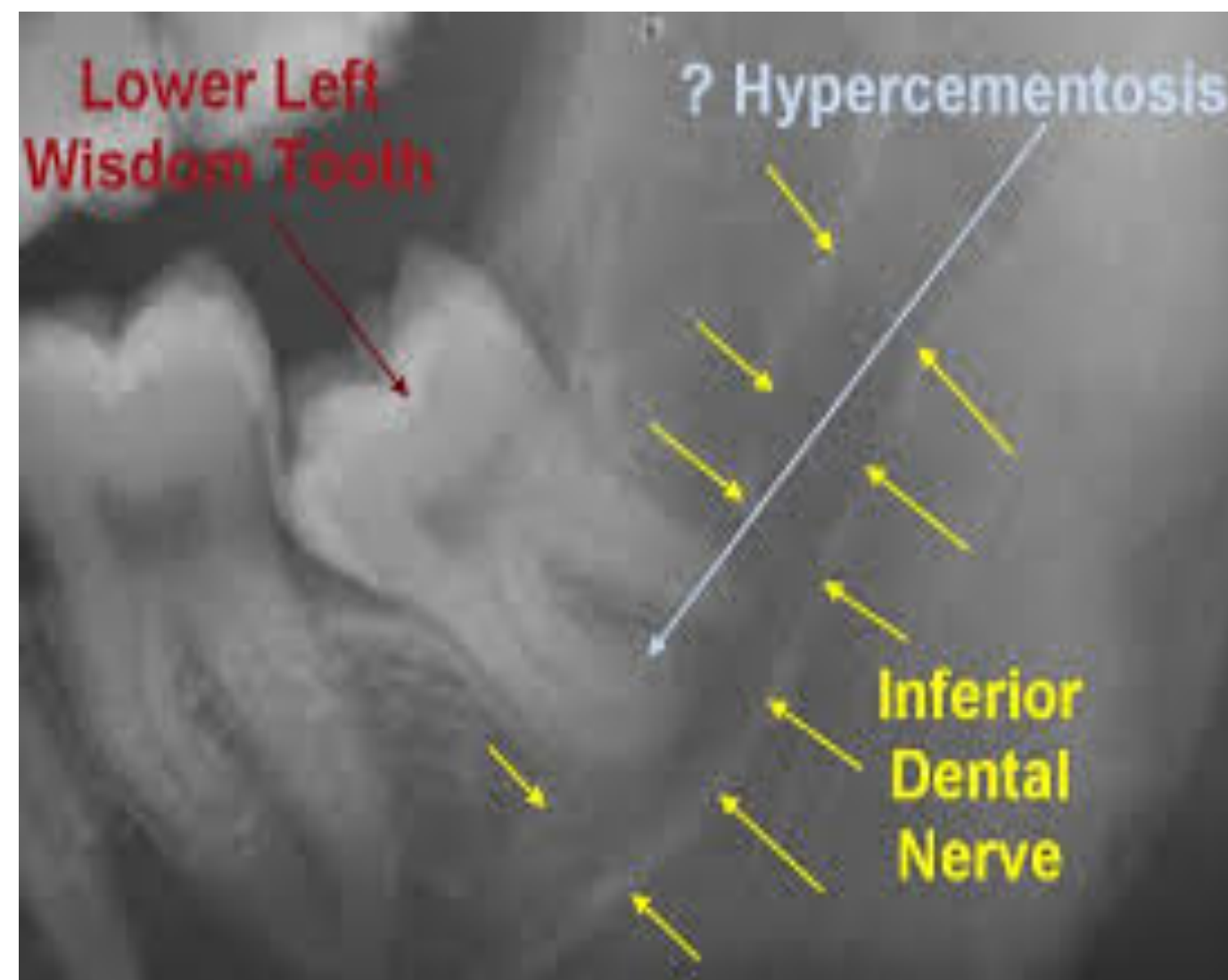
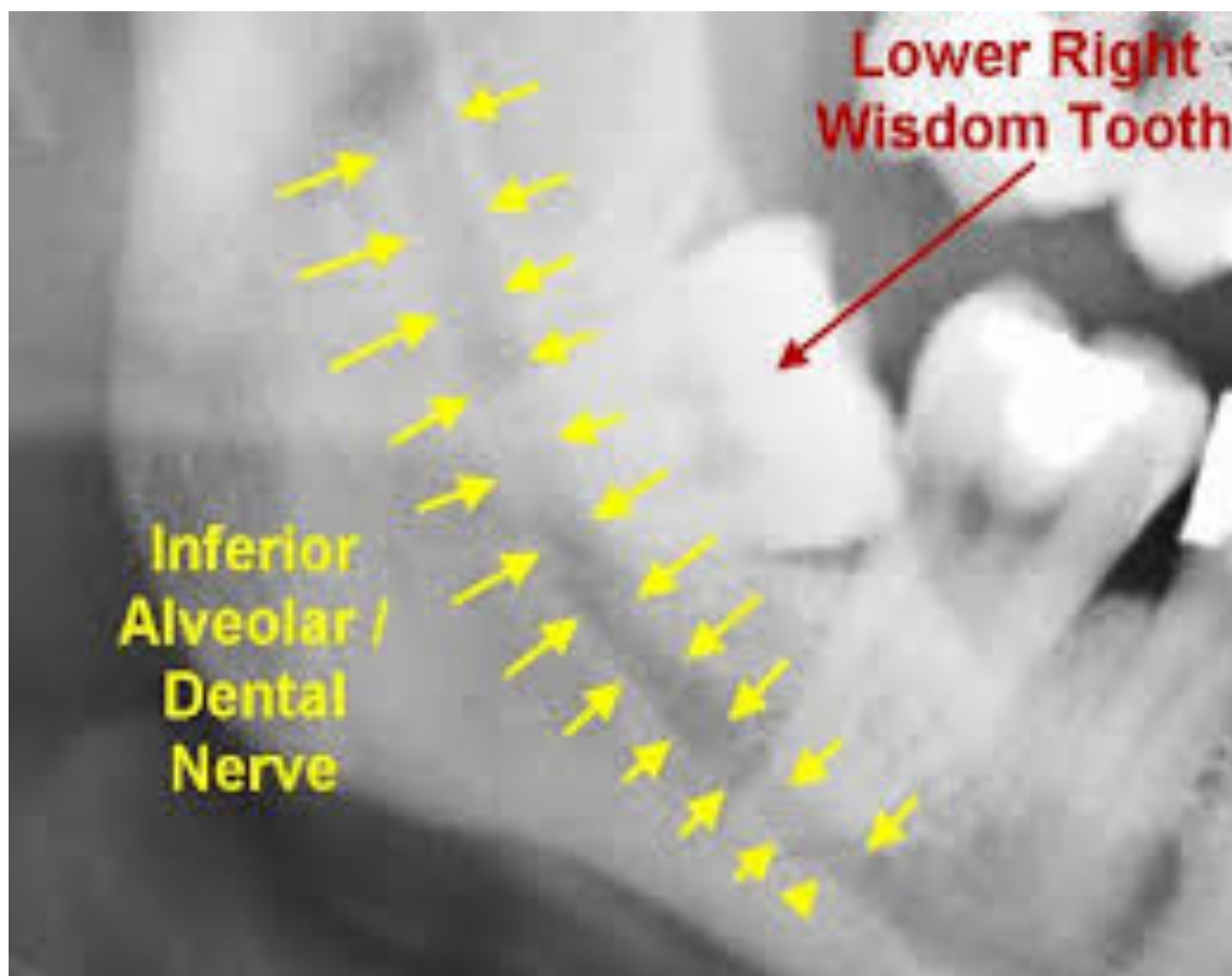
- LOWER LIP MAY BE CRUSHED BETWEEN THE HANDLES OF THE FORCEPS.
- LIPS MAY BE BURNED IF INSTRUMENT ARE NOT ALLOWED TO COOL BEFORE USE AFTER BEING STERILIZED.



C. Damage to nerves

occur mostly on surgical extraction of teeth rather than simple extraction but one must always be aware of the risk when operating in the region of the (inferior dental nerve, lingual nerve & mental nerve).

Inferior alveolar nerve injury is an uncommon occurrence in extraction of erupted mandibular teeth. In rare cases third molar roots may encircle the nerve that extraction of the tooth will cause nerve injury also improper use of elevators to remove root apices may cause tearing or displace bone fragments so that will impinging or pressing the nerve in the canal "inferior dental canal"



- The mental branch of the alveolar nerve also may be injured during surgical procedures in the premolar region.
- The lingual nerve may be damaged during exodontias of the lower molar teeth especially the lower wisdom tooth by trapping the lingual soft tissue in the forceps beaks or by direct trauma from misusing of elevator or by using surgical extraction to remove impacted wisdom tooth.

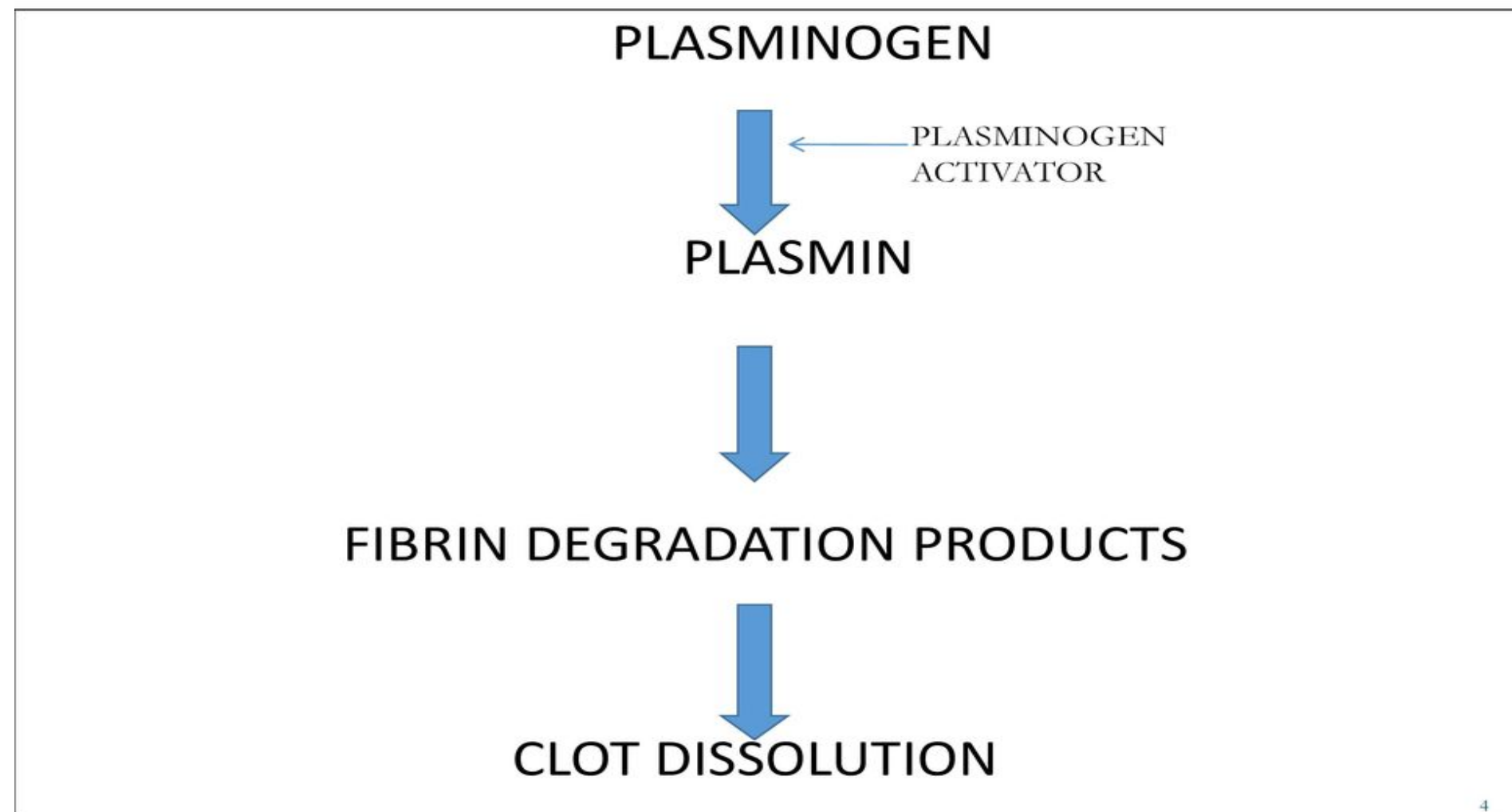
8- Dry socket

- DRY SOCKET, painful socket; alveolagia , alveolar osteitis, alveolitis sicca dolorosa or localized osteomyelitis all synonyms describing the painful condition that occurs when a blood clot at the site of tooth extraction becomes dislodged and exposed the underlying bone.
- Its incidence is approximately 3% for all routine extractions and can reach over 30% for impacted mandibular third molars.

Etiology:

- Birn's Hypothesis of dry socket: Most accepted explanation of dry socket till date (**trauma and inflammation** causes release of stable tissue activator from the adjacent bony socket and soft-tissues). Tissue activator converts plasminogen (present in the blood clot) to the plasmin. This **plasmin causes lysis of blood clot and pain by conversion of kininogen to kinin.**

THROMBOLYSIS



Predisposing factors

1. Infection of socket : release of plasminogen activators .
2. Trauma to socket and surrounding bone and soft tissue especially using burs and repeated application of elevators.
3. The use of high concentration of vasoconstrictors in the LA used for infiltration, may prevent quick formation of blood clot.
4. Mandibular extractions (dense & less vascular, contaminated with food debris).
5. Patients on oral contraceptives.
6. Smoking.

7-Excessive or early use of mouth washes.

8-Curettage after extraction of acutely infected teeth.

9-Extraction of teeth in patient having systemic disease promoting to infection such as diabetes and leukemia.

Clinical features

- Patient usually presents within 2-4 days :
- Dull, to severe throbbing pain, may radiate to the ear.
- Gingival margin of the socket is swollen & red.
- Socket may be filled with food debris or a brown friable clot, on removal exposes the bare bone which is severely tender to touch.
- Regional lymph nodes may be tender.
- Fever in some cases.
- Symptoms may last for 7- 10 days or even to 40 days.



Dry socket

Management

- 1- A radiograph should be taken to eliminate the possibility of retained root or bony fragments as a source of the pain, usually in cases when a new patient presents with such symptoms.
- 2-The socket should be irrigated with warm normal saline and all degenerated blood-clot is removed.

Management

- 3-Sharp bony spurs should be either excised or smoothed .
- 4-A dressing such as alveogyl applied on the socket.
- 5-Sedative and antibiotic could be prescribed.
- 6-Recall after 2-3 days and new dressing may be needed according to the pain severity.

9- Emphysema

- It is a swelling due to accumulation of air into the connective tissue of intermuscular facial planes .
- It usually follows prolonged reflection of large flaps together with the use of drills or air syringes, the swelling is usually very rapid in onset, giving the affected area elastic consistency.

- The condition is alarming to the patient but usually not dangerous. It take 7-10 days to be absorbed.

The dentist might prescribe rest, avoid air pressure in the area, and sometimes give antibiotics to prevent infection. In rare cases, if it spreads, more specialized care might be needed.

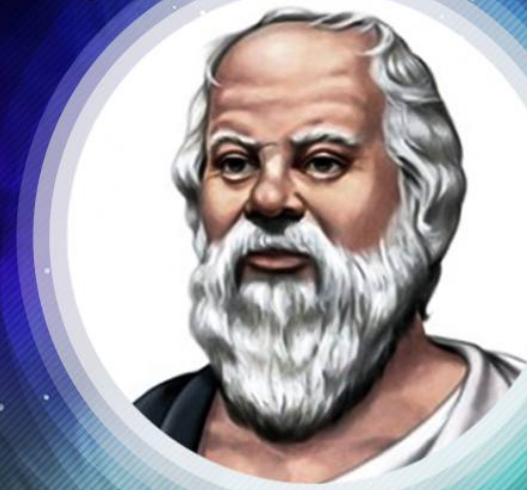
Complications could involve **swelling, difficulty swallowing, or, in extreme cases, air traveling to the chest, which can be serious.**





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